A new binary descriptor for multispectral images applied to the global localization of UAVs

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***Abstract*— This article presents a new binary descriptor for multispectral images, applied to global localization and tracking of an Unmanned Aerial Vehicle (UAV) over visible spectra colored satellite images from high vegetation areas. We propose a new measurement model based on a new version of the BRIEF descriptor and apply it in a Monte Carlo Localization system that estimates the UAV pose in 4 degrees of freedom. The method was validated using real flights, performed in two places with medium to a high amount of vegetation, and has achieved good results using high-resolution satellite images from different years.**

***Index Terms*— UAV, localization, λ-BRIEF, multispectral, satellite images, vegetation index**

I. Introduction

In the last years, there are remarkable efforts in developing aerial robots, such as multi-rotors, which have excellent maneuverability at low cost. These devices are suitable for performing tasks such as data and image acquisition, map building, and others, without dealing with terrain-related problems [1]. Among the many applications of these robots are tasks that usually required high-cost hardware and software solutions, such as precision agriculture [2].

Independent of how the robot interacts with its surrounding, it is always required to estimate the robot's pose in the environment. Although many UAVs use the Global Positioning System (GPS) to estimate their localization, this system is vulnerable [3]. Satellite obstruction is one of the problems that lead to vulnerability, once the GPS position estimation depends on the number of satellites communicating with the GPS embedded in the UAV [4]. According to Conte and Doherty [5] and Viswanathan et al. [6], devices that neutralize the GPS signal are easy to found, and their use may make unreliable the robot estimated pose. Besides, Horton and Ranganathan [7] show an apparatus for GPS spoofing and the steps to implement it using a cheap setup and open source tools, being tested with a smartphone and the DJI Matrice 100 UAV.

UAV vision-based localization systems are an excellent alternative to GPS-based systems. In these systems, the images took by the embedded camera are used by the localization system to determine the pose of the UAV regarding a known map [8]. Besides that, high-resolution satellite images can be used as a map, like in [8]–[10], since they cover most of the planet and can be obtained free of charge on platforms such as Google Earth™.

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This work was carried out with support from CNPq, National Council for Scientific and Technological Development - Brazil, and financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001.

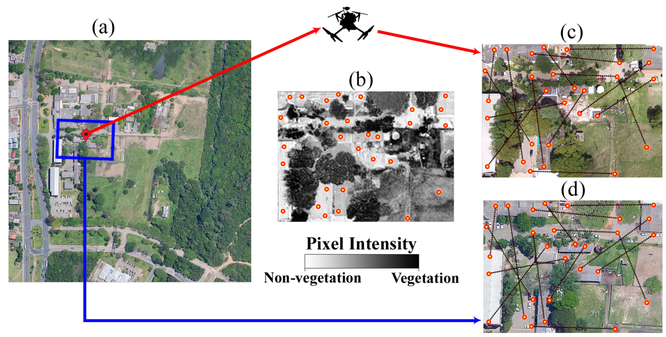

Fig. 1: A multispectral camera embedded in the UAV captures the RGB image (c) and the multispectral images used in the generation of the vegetation mask (b). The mask is used in the construction of the descriptor, selecting regions with lower vegetation density, both in the view of the UAV (c) and in the map fragment (d) from the global map (a), acting as a visualization of the particle.

Nonetheless, localization purely based on comparing images obtained by a UAV and by satellite is a challenging problem. Not only are there, in general, significant differences between the images in terms of color, lighting, and resolution, but often there are physical differences in the environment, given that the images may be obtained at very different times. Recently, Mantelli et al. [8] presented a technique using a global image descriptor called abBRIEF, that obtained good results for a drone flying over urban terrain. Still, their approach shows some problems dealing with homogeneous areas, like highly vegetated regions, that can also have significant visual changes in a little time band. In these situations, multispectral cameras play an important role in detecting vegetation in the regions that the UAV is flying over, providing important information about the ground cover for the localization process.

Recent works dealing with vision-based localization, using satellite images as a map, are restricted to RGB spectra [8]–[11]. In this paper, we propose working with estimated reflectance through the relation between visible spectra, Near Infrared (NIR), and Red Edge, by adding a multispectral camera. We combine this information to build a new binary descriptor (λ-BRIEF) based on abBRIEF [8]. This descriptor can differentiate vegetation from other surfaces, creating a robust global localization technique to estimate the UAV localization in case of GPS failure.

The remainder of this paper is organized as follows: in the next section we introduce some related work about UAV vision-based localization systems. Section III discuss existing approaches to do vegetation detection using multispectral data, while in Section IV we describe our proposal. Section V discuss the experiments and their validation, while at Section VI we conclude the work and show an idea of future work.

II. Related Work

There are works on developing vision-based localization systems for UAVs, using different types of maps and sensors, seeking to fulfill the global localization system role in GPS failure or GPS-denied environments. However, independent of the system's sensors, an environment map is necessary to localize the UAV. These vision-based localization systems can use images captured at nadir as a map, such as satellite images [8], [9] and georeferenced images [10], or can work with other kinds of maps, usually more specific and challenging to access. As an example of the second situation, Majdik et al. [11] propose a visual-localization system for UAVs flying in urban streets at low altitudes, where the surrounding buildings obstruct the GPS signal. It achieves this goal by computing the UAV position concerning the surrounding buildings, looking for correspondences between images captured by a camera on the UAV, and views generated from 3D textured models built with geotagged Google Street View™ images projected in a high-resolution 3D model from the city.

Among the works that use satellite images as a map, we highlight the approach of Masselli et al. [9], which adopted an observation model based on terrain classification with random forest trained using supervised learning. In Yol et al. [10], is presented a template-based method, which measures the similarity between the image taken by the UAV and a mosaic of georeferenced images. Moreover, the flight was done at approximately 150 m, to be able to consider the scene almost planar. In addition to these, in Mantelli et al. [8], a global vision-based localization system is presented, which works by searching for correspondences of images captured by the UAV camera in satellite images. They propose the abBRIEF descriptor, a BRIEF descriptor [12] version that also uses the color information of images. The UAV position is estimated using the Monte Carlo Localization (MCL) [13] algorithm, and the UAV odometry is obtained with feature-matching.

The recent introduction of multispectral cameras in robots, such as Parrot Sequoia, has led to the development of UAV applications that work not only with the RGB spectra information but also with the NIR spectrum. By combining them, we can estimate the chlorophyll levels of plants [14], and estimate the vegetation density of a region. After detecting regions with high vegetation density, one can assign less importance or even disregard them in the localization system, since they are regions of great uncertainty.

III. Vegetation Detection

The research in the detection and classification of vegetated areas is essential, mainly in remote sensing. Some of their applications are vegetation health analysis, cartography, and aid to autonomous ground vehicles [15]. There are many techniques for detecting chlorophyll through multispectral images [16]–[18], used to compute vegetation indices (VI).

The vegetation detection with VIs is performed based on the relation between the different wavelengths and the amount of reflectance of these wavelengths on different ground coverage, like soil, human buildings, and vegetation. It is an important data source from the ground map, providing much more information about the ground coverage than the standard natural-color image. Among these indexes, one of the most well-known and used is the Normalized Difference Vegetation Index (NDVI) [16].

Some of the well-known and most used VIs work with the NIR and red spectra, like the NDVI. The problem is that working only with these spectra, the NDVI tends to saturate in regions with medium to a high amount of vegetation [17]. One approach to threat the saturation problem is the use of the Red Edge spectrum, instead of the red from the NDVI like the Normalized Difference Red Edge Index (NDRE) [18]; or by adding the red edge spectrum to the NIR and the red from the NDVI, like in the Red and Red Edge NDVI ($NDVI_{red\&RE}$) [17].

There are works on building detection where the VI is used to build a Vegetation Mask. In Z. Zhao et al. [19], a Digital Elevation Model is built based on the LiDAR data, and an NDVI Vegetation Mask is used to help disambiguate the high areas that can be either a tree or a building. In David [15], an NDVI mask is used to differentiate the vegetation by which the vehicle crosses, from stones and other rigid obstacles.

IV. UAV Localization using natural color satellite images and UAV multispectral images

This work proposes a novel UAV global localization system that obtains images from a downward-facing multispectral camera and searches for correspondences on a map that consists of a natural-color satellite image (Fig. 1a). The localization, made with an MCL strategy, is based on a measurement model that uses a novel descriptor, called λ-BRIEF, which works with the information of invisible spectra, captured with a multispectral camera.

Our novel descriptor called λ-BRIEF is an extension of abBRIEF [8], a binary descriptor that can deal with little color changes and scene illumination. We add a VI-based vegetation mask (Fig. 1b) to explore the vegetation information available in the multispectral images. The main characteristics of λ-BRIEF are:

- the use of multispectral information in addition to color information.
- a vegetation mask computed for each pixel, responsible for determining pixel relevance, in the process o selecting pairs of pixel positions for computing the descriptor.

The global localization process requires a good likelihood measure of how similar are the actual observations made by the robot (in our case, the UAV image) and the estimated observations obtained from the known map (i.e., the satellite image). In this work, we define the measurement model as the conditional probability distribution, $p(\mathbf{d} \mid \mathbf{x}, \mathbf{m})$, of the λ-BRIEF $\mathbf{d}$, given the the robot pose $\mathbf{x}$ and the environment map $\mathbf{m}$. This probability is computed by comparing the observed λ-BRIEF descriptor by the robot, $\mathbf{d}$, with the expected $\widehat{\mathbf{d}}$ regarding the estimated pose, $\mathbf{x}$, and map, $\mathbf{m}$.

The first step of λ-BRIEF consists of computing the quantization process of the natural-color images. Secondly, a vegetation mask is computed based on a VI, which will be used in the pixel distribution. Then, pairs of pixel positions are sampled, with likelihood based on the vegetation mask, and the descriptor of the whole image is computed.

The remaining of the localization system follows Mantelli et al. [8], with the odometry obtained from feature-matching. Our code is available online[1].

A. Computing the vegetation mask - $\boldsymbol{\vartheta}$

As in [15], we work with a vegetation mask created from a VI, such as NDVI. The vegetation mask is computed for each set of drone images to identify the regions with vegetation and their concentration. Every second the camera on the drone saves a set of four multispectral images (Near-Infrared $\mathbf{I}^n$, Red Edge $\mathbf{I}^e$, Red $\mathbf{I}^r$ and Green $\mathbf{I}^g$), plus an RGB image. These multispectral images are subjected to the calculation of a VI, which will return a value between -1 and 1, where the higher the value, the greater the chances of vegetation in the pixel. The result is equalized to enhance the differences between regions without vegetation and with much vegetation.

B. Computing the gaussian mask - $\boldsymbol{\mathcal{N}}$

In addition to the vegetation mask, we also consider a gaussian mask for the pixel distribution, following the same criteria of the original BRIEF descriptor [12], prioritizing the center of the image by distributing the pairs of pixels based on two Gaussian directions starting from the center of the image. This Gaussian mask $\boldsymbol{\mathcal{N}}$ represents this same distribution, with with $\sigma_x = \frac{a}{5}$ e $\sigma_y = \frac{b}{5}$, where a is the width and b is the height from the drone image I. $\boldsymbol{\mathcal{N}}$ only needs to be computed once, as it will be the same for all images.

C. Pixel Selection Method

The pixel distribution is based on both the Gaussian mask and the vegetation mask, varying the masks' relevance. The overlay of the masks, which will be used as a basis for distributing the pixels, is given by:

$$\widehat{\boldsymbol{\vartheta}}_{\mathbf{y}_i} = (\alpha \cdot \boldsymbol{\vartheta}_{\mathbf{y}_i}) + (1 - \alpha) \cdot \boldsymbol{\mathcal{N}}_{\mathbf{y}_i} \tag{1}$$

where $\widehat{\boldsymbol{\vartheta}}_{\mathbf{y_i}}$ is a composition of the vegetation mask $\boldsymbol{\vartheta}_{\mathbf{y_i}}$ and the Gaussian mask $\boldsymbol{\mathcal{N}}$ in pixel $\mathbf{y}_i$; with α as a parameter to control the relevance of each mask. $\widehat{\boldsymbol{\vartheta}}$ will be used to set the likelihood of selecting $\mathbf{y_i}$.

So we apply the Roulette-wheel selection via stochastic acceptance method [20], a selection operator of genetic algorithms, which will be responsible for prioritizing the selection of pixels with less vegetation. The Roulette-wheel result will be $\mathbf{K}$, a set of k pairs of pixel positions, $\mathbf{p}_i = (\mathbf{y}_{i,1}, \mathbf{y}_{i,2})$ with $1 \leq i \leq k$, where each pixel position $\mathbf{p}_i$ was selected according to the probability given by Eq. 1, based on the vegetation mask $\boldsymbol{\vartheta}_{\mathbf{y}_i}$ and Gaussian mask $\boldsymbol{\mathcal{N}}$.

[1]Code: https://github.com/rswesthauser/Lambdabrief-Localization

D. Computing the observed λ-BRIEF descriptor - d

The computation of the observed λ-BRIEF considers a UAV flying parallel to the ground, only with variation in yaw. Despite our UAV not being equipped with a stabilizer gimbal, we still consider the roll and pitch angles close to zero.

During the flight, a multispectral camera directed to the ground capture a natural color image $\mathbf{I}$, and also multispectral images in near-infrared $\mathbf{I}^n$, red $\mathbf{I}^r$, red edge $\mathbf{I}^e$ and green $\mathbf{I}^g$ spectra. The environment map $\mathbf{m}$ corresponds to a large satellite image in natural color, covering the region where the robot is flying.

The observed λ-BRIEF descriptor $\mathbf{d}$ is created based on the image $\mathbf{I}$, being composed by binary vectors, where each vector $\mathbf{d}_i = (\tau_{i,1} ... \tau_{i,c})^T$ is associated to the pair of pixel positions $\mathbf{p}_i$ and c is the number of image color channels. Each element $\tau_{i,c}$ is the result of a binary comparison between the j^{th} color channel at pixel position $\mathbf{y}$ from image $\mathbf{I}$, $\eth^j(\mathbf{y}_{i,1})$ and $\eth^j(\mathbf{y}_{i,2})$, from the j^{th} color channel of pixels at positions $\mathbf{y}_{i,1}$ and $\mathbf{y}_{i,2}$ of the pair of pixels belonging to $\mathbf{p}_i \in \mathbf{K}$, i. e.,

$$\tau_{i,j} = \begin{cases} 1, & \text{if } \eth^j(\mathbf{y}_{i,1}) > \eth^j(\mathbf{y}_{i,2}) \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

E. Computing the expected λ-BRIEF descriptor - $\widehat{\mathbf{d}}$

Given an estimated robot pose, $\mathbf{x}$, we generate the expected λ-BRIEF descriptor $\widehat{\mathbf{d}}$ by spreading the same k pairs of pixel positions from $\mathbf{K}$ in the expected image $\widehat{\mathbf{I}}$. For that we align the pairs of pixel positions from $\mathbf{I}$ to $\widehat{\mathbf{I}}$, performing the coordinate transformation

$$\widehat{\mathbf{Y}}_{i,b} = \begin{bmatrix} zR(-\Theta) & \mathbf{y} \\ 0 & 1 \end{bmatrix} \cdot \mathbf{y}_{i,b}, \text{with } b = 1, 2, \tag{3}$$

where $\widehat{\mathbf{Y}}_{i,b}$ is the new pixel position, while $\mathbf{y}$ is the corresponding pixel position of the robot pose in $\mathbf{m}$; $\mathbf{b}$ is the pair index; $\mathbf{z}$ is a scale factor that relates the height of the robot flight and the area of $\mathbf{m}$ observed by the robot; Θ is the robot orientation and $R \in SO(2)$ is a 2D clockwise rotation matrix [8]. Then we do the same computing from the observed descriptor, but with the new pairs of pixel positions from $\widehat{\mathbf{K}}$.

F. Computing the similarity between the observed and expected descriptors

The UAV and satellite natural color images $\mathbf{I}$ and map $\mathbf{m}$ must be in the same color space and preprocessed before the descriptor computation. The similarity between the observed $\mathbf{d}$ and expected $\widehat{\mathbf{d}}$ descriptors is given by the comparison between the descriptors $\mathbf{d}$ and $\widehat{\mathbf{d}}$. The process of computing the similarity between the observed and the expected descriptors, $\mathbf{d}$ and $\widehat{\mathbf{d}}$, is defined as

$$\xi(\mathbf{d}, \widehat{\mathbf{d}}) = \eta \sum_{i=1}^{k} \sum_{j=1}^{c} \tau_{i,j} \,\overline{\oplus}\, \widehat{\tau}_{i,j} \qquad (4)$$

where $f \,\overline{\oplus}\, g$ is the XNOR logic operator, $\eta = (kc)^{-1}$ and c is the number of channels of the color space used. In our case c = 2 because we are working with the color channels of the CIE *L*a*b** color space, without the lightness channel *L**.

V. Experiments

All flights were performed by a DJI Matrice 100 equipped with a downward-looking camera, with a fixed mounting. The flying was captured at 1 FPS, with pixel dimensions of 1280×960 for the multispectral, and 4608×3456 for the natural color images.

The experiments were computed on a desktop AMD FX 6300 with six cores of 3.5 GHz and 16 GB RAM. All multispectral images were preprocessed, removing the fisheye distortion based on the procedure described on the developer note provided by Parrot and rectified. Simultaneously, the natural color images were downsampled to 1280×960 and aligned with the multispectral images, while all maps have a resolution of 4800×4800.

TABLE I: Flights

Name	Length (m)	Number of images	Altitude (m)
UFRGS	826	224	82-101
ADM1	1169	300	60-191
ADM2	716	196	79-103
ADM3	1610	266	32-169

Four flights were performed in Brazil, the first[2] over the city of Porto Alegre, more specifically, over the campus of the Federal University of Rio Grande do Sul (UFRGS) on 05-05-2018, while the other three (**ADM1**[3], **ADM2**[4] and **ADM3**[5]) were performed over the city of Arroio do Meio on 04-18-2019. The detailed information about the flights is available in Table I.

For the flight carried out at UFRGS, twelve satellite images were extracted from Google Earth™ [6], with an area of approximately 0.48 km^2, seeking to cover a wide range of seasons and years. The maps from Arroio do Meio have approximately 0.9 km^2, also covering a wide range of seasons and years.

All experiments were carried out by varying the values of α in steps of 0.25, ranging from 0 to 1, where 0 indicates the total absence of vegetation information in the process. Therefore, with $\alpha = 0$, the distribution of the pixel pairs on the image is the same used by abBRIEF [8].

[2]Video youtu.be/TnzhlY7X3BM and dataset zenodo.org/record/4076075.
[3]Video youtu.be/wPCHsxEqplo and dataset zenodo.org/record/4067437.
[4]Video youtu.be/zn4e4oAmpEE and dataset zenodo.org/record/4068506.
[5]Video youtu.be/Gwhl8xtKc3k and dataset zenodo.org/record/4070013.
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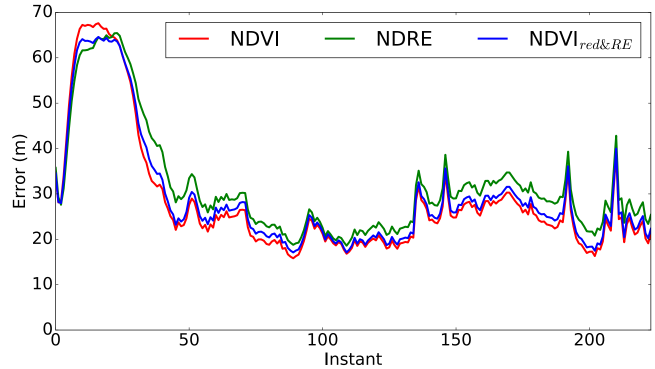


Fig. 2: Comparison of errors obtained by selecting different vegetation indices in experiments with the UFRGS trajectory.

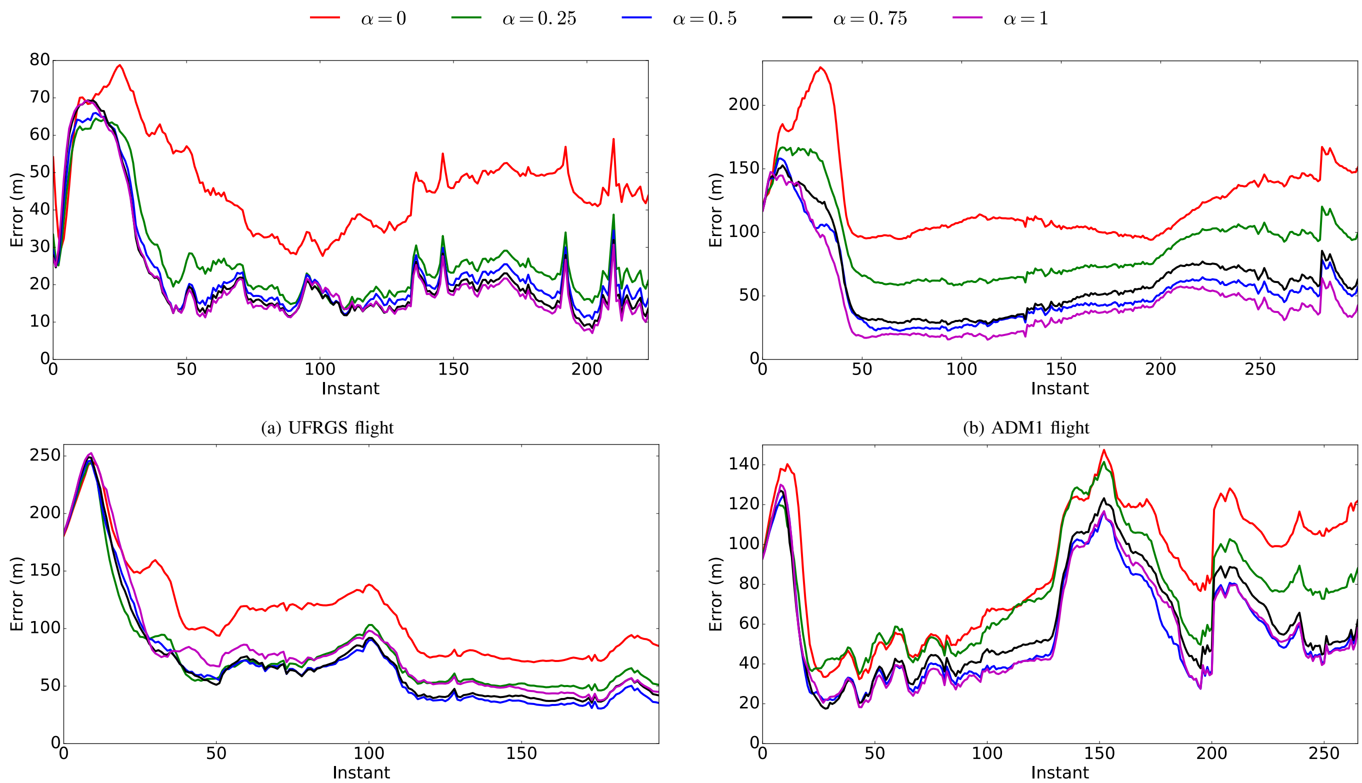
The MCL estimate was compared with the GPS data which was considered the Ground truth. The particles are initially spread evenly on the satellite image map considering the orientation $\Theta \in [0, 2\pi)$ and the altitude $z \in [20$ m, 250 m] for the trajectories of Arroio do Meio, and $z \in [46$ m, 466 m] for the trajectory of UFRGS. At each instant t of the trajectory the estimate of the position of the UAV is given by $\bar{x}_t = A^{-1} \sum_{i=1}^{n} w_t^{[i]} \mathbf{x}_t^{[i]}$ [8], where n is the number of particles, $w_t^{[i]}$ the weight and $\mathbf{x}_t^{[i]}$ the pose of the ith particle, and $A = \sum_{i=1}^{n} w_t^{[i]}$.

To obtain a statistically significant estimate, for each combination of α and $\mathbf{m}$, the experiment was repeated $\gamma = 30$ times. Moreover, for position estimation using the MCL measurement model based on λ-BRIEF 50.000 particles, and 256 pairs of pixel positions were used, both to abBRIEF and λ-BRIEF. So the position estimate error ε_t is given by the average difference between the GPS data $\mathbf{x}_t^r = (x_t^r, y_t^r, z_t^r)$ and the position $\bar{x}_t$, at instant t, $\gamma^{-1} \sum_{j=1}^{\gamma} \sqrt{(\mathbf{x}_t^r - \bar{\mathbf{x}}_t^j)^2}$ [8]. In order to calculate the error of the angle estimate, the average angle $\bar{\Theta}_t$ of all n particles is first calculated $\bar{\Theta}_t = \arctan(\sum_{i=1}^{n} \sin(\Theta_t^{[i]}) w_t^{[i]}, \sum_{i=1}^{n} \cos(\Theta_t^{[i]}) w_t^{[i]})$, where $w_t^{[i]}$ is the weight and $\Theta_t^{[i]}$ is the orientation of the ith particle. Then the error of the angle estimate $\varkappa_t$ is given by the difference between the average error of all particles $\bar{\Theta}_t$, and the ground truth angle Θ_t^r given by the IMU sensor, $\varkappa_t = \arctan(\sum_{i=1}^{\gamma} \sin(\bar{\Theta}_t^{[i]}), \sum_{i=1}^{\gamma} \cos(\bar{\Theta}_t^{[i]})) - \Theta_t^r$.

A. Different Vegetation Indices

The definition of the VI to be used in the experiments was given based on preliminary experiments executed in the UFRGS trajectory, where three different VIs were compared. We opted for NDVI [16] for being an index widely used for vegetation detection, which works with the NIR spectrum, typically found in multispectral cameras; and two vegetation indices also using the Red Edge spectrum, the NDRE [18], and the NDVI$_{red\&RE}$ [17], in order to verify the impact of the saturation reduction provided by these indices in the process of localizing the UAV.

The results were computed for all twelve UFRGS maps, to all α values. On Fig. 2, a slightly larger error is noticed right at the beginning of the trajectory for the NDVI, but



(a) UFRGS flight

(b) ADM1 flight

(c) ADM2 flight

(d) ADM3 flight

Fig. 3: Experiments varying α, for the flight performed at UFRGS, as well as the three flights performed at Arroio do Meio Figs. 3b, 3c, and 3d. The higher the value of α, the greater the relevance of the vegetation mask in the distribution of pixels to be used by λ-BRIEF, while for α = 0, the system behaves like abBRIEF.

as soon as the method converges, the error remains much lower when compared to the other VIs. Still, in this graph, it is apparent that the VIs that work with the NIR spectrum (NDVI, $\text{NDVI}_{\text{red\&RE}}$) showed a better performance in the localization process when compared to the VI that replaces the NIR spectrum by the Red edge (NDRE). The experiments demonstrate that NDVI is the best VI for working on λ-BRIEF, among those tested. Thus, the remaining experiments were performed using NDVI.

B. Position estimation

The four plots shown in Fig. 3 present the average error of each flight, for all five maps. In the UFRGS flight (Fig. 3a), the error decreases in proportion to the increase in the value of α. This same pattern is seen in the other trajectories, but with $\alpha > 0.5$, the result tends to be similar, as can be seen in Fig. 3b and Fig. 3d. In Figs. 3a, 3c and 3d the use of the lowest non-zero value of α already has a great influence on the result, with saturation at $\alpha > 0.5$.

The λ-BRIEF was developed to be a robust descriptor, especially when there is a high amount of vegetation. This section will present the results of varying the value of α, using the VI tested before. Moreover, it will compare between λ-BRIEF using the best value for α, with the descriptor abBRIEF, applied to estimate the localization of a UAV flying in regions with vegetation.

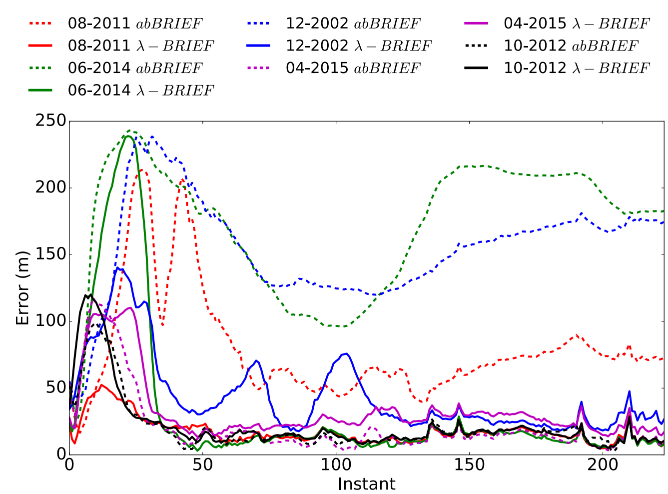
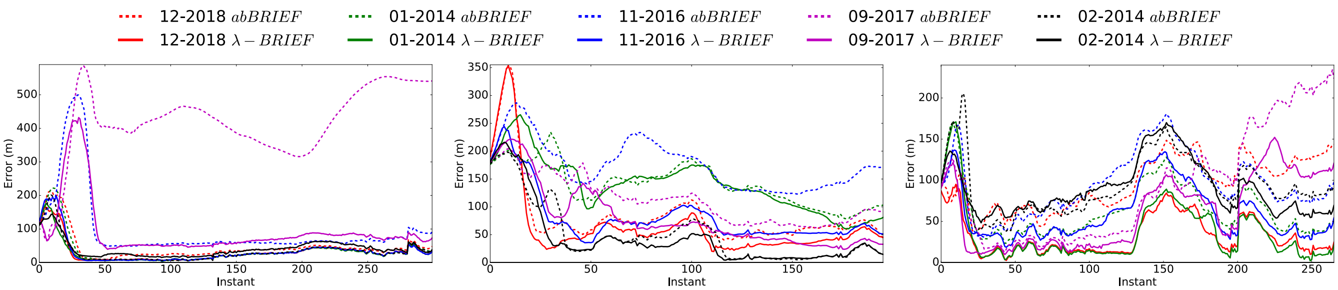


Fig. 4: Comparison between abBRIEF and λ-BRIEF fot the UFRGS Flights.

These four plots show the average error of each flight, for all five maps of Arroio do Meio and eight maps of UFRGS, listed before. Higher values of α have a smaller error, and this can be seen on almost all flights, except for ADM2 flight Fig. 3c, where all examples show similar results.

Finally, we compare our method with the value α = 1, with the approach without using vegetation information, presented in Mantelli et al. [8]. In Fig. 4, we show the results



(a) ADM1 flight

(b) ADM2 flight

(c) ADM3 flight

Fig. 5: Comparison between abBRIEF and λ-BRIEF for the Arroio do Meio Flights.

of five maps that present the most remarkable differences in the error estimate. We see that after a slightly slower convergence, our method generally shows much better results (i.e., smaller error) than the approach in [8], only on maps 04-2015 and 10-2012 there is no clear advantage, but this happens because the filter also converges using abBRIEF.

In the case of Arroio do Meio flights, we display the results for all five maps, even when very similar. Here again, there is a slightly larger error, right at the beginning of the flight, but as soon as the method converges, the error is much smaller. In the other two flights made in Arroio do Meio, as soon as the convergence occurs, our method presents better or similar results for all five maps, except in the ADM3 flight (Fig. 5c) for the map from 02-2014.

VI. Conclusion

This work proposes a multispectral binary descriptor applied in a UAV localization method equipped with a downward multispectral camera, without a gimbal, in an outdoor environment with a high amount of vegetation. The main contributions of this article are:

- the development of a multispectral binary descriptor, capable of distinguishing the amount of vegetation on different parts of the image treating each differently.
- a vision-based UAV localization system, with 4 DoF, which can localize the UAV considering trajectories with high and medium amount of vegetation, in several different satellite image maps from different years with more than 0.48 km^2 of area, and 420 m of altitude range in the MCL search space.
- a Monte Carlo Localization approach that is fast and able to recover from localization failure. In our experiments, [8] took 4,72 min to calculate the same trajectory, while our method took 4,79 min to calculate the same trajectory, but with an additional vegetation computation and showing better results.

As future work, we intend to use multispectral satellite images, acting as a global map. With this approach, we can build a new descriptor that stores the binary comparison result from the VI computed for the local and global map, improving the localization.

References

[1] A. Ollero and L. Merino, "Control and perception techniques for aerial robotics," in *Annual Reviews in Control*, vol. 28, no. 2, 2004, pp. 167 – 178.

[2] J. Navia, I. Mondragon, D. Patino, and J. Colorado, "Multispectral mapping in agriculture: Terrain mosaic using an autonomous quadcopter uav," in *2016 International Conference on Unmanned Aircraft Systems (ICUAS)*, June 2016, pp. 1351–1358.

[3] J. V. Carroll, "Vulnerability assessment of the U.S. transportation infrastructure that relies on the global positioning system," in *The Journal of Navigation*, vol. 56, 5 2003, pp. 185–193.

[4] F. Caballero, L. Merino, J. Ferruz, and A. Ollero, "Improving vision-based planar motion estimation for unmanned aerial vehicles through online mosaicing," in *International Conference on Robotics and Automation*, May 2006, pp. 2860–2865.

[5] G. Conte and P. Doherty, "An integrated UAV navigation system based on aerial image matching," in *Aerospace Conference*, March 2008, pp. 1–10.

[6] A. Viswanathan, B. R. Pires, and D. Huber, "Vision-based robot localization across seasons and in remote locations," in *International Conference on Robotics and Automation*, May 2016, pp. 4815–4821.

[7] E. Horton and P. Ranganathan, "Development of a gps spoofing apparatus to attack a dji matrice 100 quadcopter," *The Journal of Global Positioning Systems*, vol. 16, no. 1, p. 9, Jul 2018. [Online]. Available: https://doi.org/10.1186/s41445-018-0018-3

[8] M. Mantelli, D. Pittol, R. Neuland, A. Ribacki, R. Maffei, V. Jorge, E. Prestes, and M. Kolberg, "A novel measurement model based on abbrief for global localization of a uav over satellite images," *Robotics and Autonomous Systems*, vol. 112, 12 2018.

[9] A. Masselli, R. Hanten, and A. Zell, "Localization of unmanned aerial vehicles using terrain classification from aerial images," in *Intelligent Autonomous Systems*, 2016, pp. 831–842.

[10] A. Yol, B. Delabarre, A. Dame, J. É. Dartois, and E. Marchand, "Vision-based absolute localization for unmanned aerial vehicles," in *International Conference on Intelligent Robots and Systems*, Sept 2014, pp. 3429–3434.

[11] A. Majdik, D. Verda, Y. Albers-Schoenberg, and D. Scaramuzza, "Air-ground matching: Appearance-based gps-denied urban localization of micro aerial vehicles," *Journal of Field Robotics*, vol. 32, 05 2015.

[12] M. Calonder, V. Lepetit, C. Strecha, and P. Fua, "BRIEF: Binary Robust Independent Elementary Features," in *European Conference on Computer Vision*, 2010, pp. 778–792.

[13] F. Dellaert, D. Fox, W. Burgard, and S. Thrun, "Monte carlo localization for mobile robots," in *Proceedings 1999 IEEE International Conference on Robotics and Automation (Cat. No.99CH36288C)*, vol. 2, 1999, pp. 1322–1328 vol.2.

[14] A. A. Gitelson, M. N. Merzlyak, and H. K. Lichtenthaler, "Detection of red edge position and chlorophyll content by reflectance measurements near 700 nm," *Journal of Plant Physiology*, vol. 148, no. 3, pp. 501 – 508, 1996. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0176161796802859

[15] D. M. Bradley, R. Unnikrishnan, and J. Bagnell, "Vegetation detection for driving in complex environments," in *Proceedings 2007 IEEE International Conference on Robotics and Automation*, April 2007, pp. 503–508.

[16] J. Rouse, J. W., "Monitoring the vernal advancement and retrogradation (green wave effect) of natural vegetation," Texas A&M Univ.; Remote Sensing Center.; College Station, TX, United States, Tech. Rep., 01 3 1974.

[17] Q.-y. Xie, J. Dash, W. Huang, D. Peng, Q. Qin, H. Mortimer, R. Casa, S. Pignatti, G. Laneve, S. Pascucci, Y. Dong, and H. Ye, "Vegetation indices combining the red and red-edge spectral information for leaf area index retrieval," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 02 2018.

[18] E. M. Barnes, T. R. Clarke, S. E. Richards, P. D. Colaizzi, J. Haberl, M. Kostrzewski, P. Waller, C. Choi, E. Riley, T. Thompson, R. J. Lascano, H. Li, and M. S. Moran, "Coincident detection of crop water stress, nitrogen status and canopy density using ground based multispectral data," in *In Proceedings of the Fifth International Conference on Precision Agriculture*, 2000.

[19] Z. Zhao, H. Wang, C. Wang, S. Wang, and Y. Li, "Fusing lidar data and aerial imagery for building detection using a vegetation-mask-based connected filter," *IEEE Geoscience and Remote Sensing Letters*, vol. PP, pp. 1–5, 02 2019.

[20] A. Lipowski and D. Lipowska, "Roulette-wheel selection via stochastic acceptance," *Physica A: Statistical Mechanics and its Applications*, vol. 391, no. 6, pp. 2193 – 2196, 2012. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0378437111009010